SSH BRUTE FORCE LAB - KEY FINDINGS ANALYSIS

ATTACK SUMMARY

Type: SSH Brute Force + Post-Compromise Reconnaissance

Duration: ~15-20 seconds

Attacker: 192.168.56.103 (Kali Linux)

Victim: 192.168.56.102 (Windows 11)

Target Account: vbox

Result: SUCCESSFUL COMPROMISE

DETECTION METRICS

FAILED LOGIN ATTEMPTS

- Count: 5
- Event ID: 4625
- Source: All from 192.168.56.103
- Status Code: 0xC000006D (bad password)

SUCCESSFUL LOGIN

- Count: 1
- Event ID: 4624
- Source: 192.168.56.103
- Status Code: 0x0 (success)
- Logon Type: 3 (Network/SSH)

POST-COMPROMISE COMMANDS

- Count: 3
- Event ID: 4688 (Process Creation)

- Commands:
  1. whoami - Confirm identity
  2. net user - Enumerate users
  3. ipconfig - Map network

ATTACK PROGRESSION

PHASE 1: RECONNAISSANCE (T+0 to T+10 seconds)

Attacker guesses passwords systematically.

Indicators:

- 5 x 4625 events (failed logins)
- Same account, same IP, rapid succession
- Severity: HIGH (brute force pattern detected)

PHASE 2: ACCESS (T+12 seconds)

Attacker gains access with correct password.

Indicators:

- 1 x 4624 event (successful login)
- Same source IP, same account
- Severity: CRITICAL (account compromised)

PHASE 3: RECONNAISSANCE (T+13 to T+17 seconds)

Attacker gathers system information.

Indicators:

- 3 x 4688 events (process execution)

- Enumeration commands executed

- Severity: CRITICAL (active attacker on system)

CORRELATION ANALYSIS

TIMELINE ALIGNMENT

Kali (Attacker): Windows (Victim):

19:45:30 Start ---

19:45:32 wrong1 -----> 4625 Failed

19:45:34 wrong2 -----> 4625 Failed

19:45:36 wrong3 -----> 4625 Failed

19:45:38 wrong4 -----> 4625 Failed

19:45:40 wrong5 -----> 4625 Failed

19:45:42 SUCCESS -----> 4624 Success

19:45:43 whoami -----> 4688 Process

19:45:45 net user ----> 4688 Process

19:45:47 ipconfig ----> 4688 Process

Result: Perfect timeline correlation confirms single attack campaign

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RISK ASSESSMENT

Component Risk Level Justification

Account Access 🔴 CRITICAL Account compromised

System Information Leakage 🔴 CRITICAL Network & users exposed

Privilege Escalation Risk 🔴 HIGH Attacker has foothold

Lateral Movement Risk 🔴 HIGH Network mapping done

Data Exfiltration Risk ⚠️ MEDIUM Depends on next steps

IMMEDIATE ACTIONS REQUIRED

1. LOCK/DISABLE ACCOUNT
   - Prevent further login
   - Force password change

2. ISOLATE SYSTEM
   - Disconnect from network if possible
   - Prevent lateral movement

3. PRESERVE EVIDENCE
   - Export all security logs
   - Create full system backup
   - Document timeline

4. INVESTIGATE FURTHER
   - Check for additional compromise
   - Review other accounts for unauthorized access
   - Check for backdoors or persistence mechanisms

DETECTION QUALITY ASSESSMENT

WHAT WORKED:

✓ Event logging captured all activities

✓ Failed and successful logins logged

✓ Process creation logging showed commands

✓ Rapid detection was possible

IMPROVEMENT OPPORTUNITIES:

- Account lockout policy not enabled (prevented self-remediation)
- No real-time alerting (detection was manual)
- No MFA (weak single-factor auth)
- Password-based auth allowed (best practice: use keys)

LESSONS FOR BLUE TEAM

1. LOG EVERYTHING

   You captured the complete attack

2. CORRELATE TIMELINES

   Multiple logs told same story

3. ACT FAST

   Account lockout would have stopped attack

4. MFA MATTERS

   Password alone insufficient

5. MONITOR COMMANDS

4688 revealed attacker intent

KEY TAKEAWAYS

For Security Professionals:

- Brute force attacks create obvious patterns
- Log correlation reveals complete incidents
- Process execution logging is critical
- Cross-platform analysis is powerful
- Rapid response prevents escalation

For Lab Learners:

- Understand attack lifecycle
- Know what logs to look for
- Learn to correlate multiple events
- Practice incident analysis
- Document findings professionally

Evidence Quality: ☆ ☆ ☆ ☆ ☆ (5/5)

Perfect correlation between attacker and victim logs

All attack phases documented with timestamps

Clear indicators of compromise

Professional-grade analysis
